

Will Quantum Computers Transform Our Lives More Than Classical Ones?

2 of 5: Materials Science, Global Startups, and the Governance Challenge



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Thoughts on Quantum Technology Based on Interpretations at SEEQA – Quantum Conference Hosted at Oxford University

Every technology that has redefined civilization—bronze, steel, silicon—has begun with materials. What we build with shapes is what we build toward. Classical computing gave us semiconductors, polymers, and superalloys that powered the 20th century. But when it comes to simulating matter itself—its electrons, atoms, and their delicate dance—classical machines hit a wall. Quantum computers, by contrast, speak the native language of matter. That's why their impact on **materials science** could ultimately eclipse the achievements of conventional computing.

Where Classical Simulation Falls Short

Materials behave according to quantum mechanics, yet classical supercomputers must approximate these interactions. Each additional qubit exponentially increases complexity. That's why it still takes decades to go from discovery to deployment in fields like battery chemistry or superconductivity. We are running 21st-century industries on 20th-century trial-and-error.

Quantum Simulation: A Natural Fit

Quantum computers encode molecular interactions directly into qubit today's modest devices have crossed benchmarks that classical system cannot. In 2025, researchers paired a 77-qubit processor with a classic supercomputer to calculate the properties of iron-sulphur clusters—molecules already beyond exact classical simulation. As error rates fall qubit counts rise, this hybrid approach will expand into industrially rel problems:

- **Next-generation batteries** with higher energy density and faster charging.
- **Catalysts rivalling natural enzymes** for ammonia synthesis, cutting enormous energy cost of fertilizer production.
- **Room-temperature superconductors** that could transform power transport, and medical imaging.

These aren't abstract promises. They are grand challenges with measu economic stakes.

A Global Startup Wave

Momentum is not confined to labs. Startups across continents are turing quantum materials research into platforms, partnerships, and product

- **HQS Quantum Simulations** (Germany) develops quantum algorithms to predict properties of specialty materials.
- **Ergo Quantum** (Canada) integrates classical and quantum models for

material design.

- **Multiverse Computing** (Spain) offers Singularity, a platform where engineers can apply advanced algorithms without deep quantum t
- **IQM** (Finland/Germany) leads the QuantiCoM project on quantum simulation for materials, backed by national aerospace funding.
- **XtalPi** (China) extends its AI-quantum hybrid approach from pharmaceuticals into solar and EV battery materials.
- **Xanadu** (Canada) advances photonic quantum hardware with cloud accessible tools for chemistry and materials.
- **Alice & Bob** (France/USA) build “cat-qubit” fault-tolerant systems, for scalable simulations.
- **Atom Computing** (USA) pushes toward thousand-qubit neutral atom machines that could anchor large-scale quantum chemistry.

These ventures reflect a distributed innovation ecosystem: Europe emphasizes industrial partnerships; North America leans into platform scalability; Asia ties quantum directly to strategic industries like energy mobility.

The Governance Angle

This is not just a technical story—it’s a governance story. If quantum computing unlocks the chemistry of matter, committees and boards will have to ask: *who sets the rules, who captures the value, and who bears the risk*

- **Audit & Risk Committees:** Treat quantum-enabled R&D as a new category of strategic risk. Dependencies on rare earths, photonics supply chain

and proprietary platforms could create geopolitical vulnerabilities.

- **Science & Technology Committees:** Push for reproducibility and benchmarking. Without standards, “quantum advantage” risks being marketing rather than science. Supporting cross-industry consortia (like QuPharm or QED-C) is essential.
- **Sustainability Committees:** Quantum-accelerated discoveries could make fertilizer or batteries more sustainable—but governance must ensure environmental and social externalities are addressed. Who gains access first: multinationals or smallholders?
- **Compensation Committees:** Incentives will need to evolve. How do we reward teams working on decade-long quantum research arcs, where commercial payoffs are uncertain but potentially transformative?

The message is clear: until fault-tolerant machines arrive, governance would benefit from reviewing their oversight frameworks, funding collaborations, and anticipating regulatory standards.

Investor and Market Signals

Capital markets are already validating the thesis. Startups like **Quantum Machines** have raised over \$170 million in late-stage funding ([WSJ](#)). Meanwhile, the **Financial Times** frames software as the “next frontier” in quantum computing, underscoring that the race is shifting from raw hardware to chemistry-driven applications ([FT](#)). For boards, the governance question is not *if* quantum enters the enterprise, but *when and on whose terms*.

Balanced Expectations

Realism is crucial. Devices remain noisy and limited, fault-tolerance is years away, and industrial workflows will demand careful governance. trajectory is undeniable: AI accelerates the search space, quantum provides the fidelity, and human expertise sets direction.

Final Thought

For most of us, the impact of quantum computing in materials science will be felt in daily life. Imagine phone batteries that last for days instead of hours or electric cars that charge in minutes rather than overnight. Fertilizers with far less energy could bring down food prices and reduce pressure on the planet. Power grids could become more stable, lowering the chance of blackouts during heatwaves or winter storms. Even medical devices like MRI machines could become cheaper and more widely available if superconducting materials improve. In short, while the research may seem abstract, the result could be cleaner energy, more affordable food, and better healthcare—tangible changes that shape the rhythm of everyday life.

If classical computing was the engine of the digital revolution, quantum computing will be the forge of the material revolution. By enabling us to simulate matter at the deepest level, quantum computers could unlock batteries that stabilize the grid, fertilizers that feed billions sustainably, and superconductors that reduce energy loss in infrastructure.

For governance leaders, keep watching the space!

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